# Pop Song English and Singing Technique:

# Investigating the Role of Formant-Harmonic Alignment

Andy Gibson, University of Oxford

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## Sociophonetics of pop music: Stylistic norms and singing technique

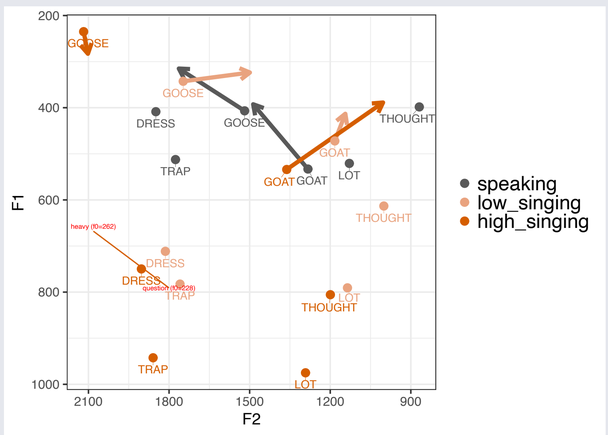



Figure 1. Dylan Storey's mean vowel measures (n=277) in speech (New Zealand English), low singing (f0 overlaps with speech f0 range) and high singing (Gibson & Bell 2012)

### Pop Song English (PSE)

- American-influenced supralocal norm (Gibson 2024)
- Emerged through acts of identity and differs by genre (Trudgill 1983)
- Maintained through context-dependent associative learning (Gibson & Hay 2026)
- But… **singing is sung!** Additional physiological/acoustic constraints
- Fig. 1: F1 rises from speech to low singing to high singing
- Increased jaw opening at higher f0 for trained and untrained singers alike (Austin 2007)

### Formant-Harmonic Alignment (FHA)

- **Formant tuning**: align a vocal tract resonance with a harmonic to enhance amplitude or achieve desired timbre
  - Sopranos keep F1>f0 (Joliveau et al. 2004), musical-theatre belt aligns F1 with H2 (Herbst et al. 2023),
  - F1 aligns with n*f0 in clear speech (Story et al. 2025)
  - Formant tuning can 'distort' vowel quality and be '**consciously employed**' (Schutte et al. 2005: 297)
  - **BUT**: Male singers maintained vowel quality over f0 range… FHA largely "**by chance**" Sundberg et al. (2013: 287)

#### *How much formant tuning in pop?*

- Sung/spoken vowels (n=697) from 3 male NZ singers
- FHA similar in speech and song when f0 controlled, **formant tuning not a general strategy in song**
- Closer FHA on long vowels: some intentional FHA? (Gibson 2026 based on Gibson & Bell 2012 data, n=697)

***RQ: What is the role of singing-technique in speech-singing differences?***

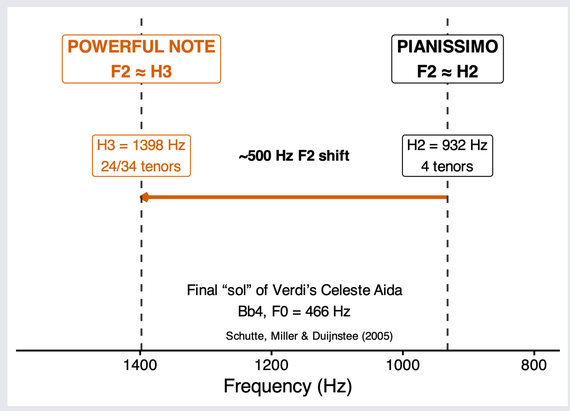



Figure 2. In classical singing, formant tuning can be dramatic. On final B♭4 of Verdi's Celeste Aida, 24/34 tenors tuned F2 to H3 for a powerful note, despite 'marked distortion' of vowel quality. 4/34 tenors used F2-H2 tuning in pianissimo (Schutte et al. 2005)

## FHA: Singing technique or measurement error?

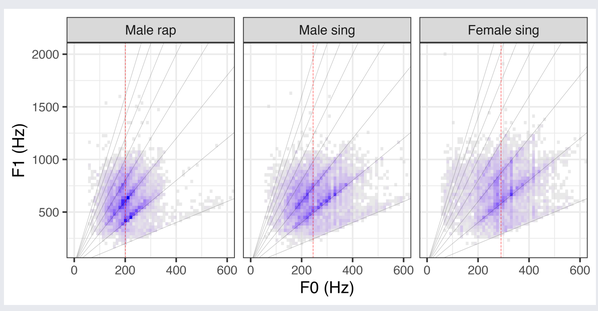



Figure 3. F1 measurements from 25,343 vowels in PoPS corpus (Gibson 2024). Measures cluster around harmonics. Apparent formant-harmonic alignment, but is this just **harmonic attraction?**

### Is this formant tuning? *(Nope)*

- Methods:
  - 190 NZ/US songs from NZ charts (Gibson 2024)
  - Vocals separated with demucs (Défossez 2021)
  - Vowels force-aligned with MFA (McAuliffe et al. 2026)
  - Formants estimated with FastTrack (Barreda 2020)
  - Median F1 from middle fifth of vowel, f0 from Reaper.
  - Singing/rap distinguished automatically with duration variability measure, uncertain tokens removed
- Many F1 measurements are near a harmonic, in both rap and singing
- Same proportion of F1 measurements close to a harmonic as Gibson (2026)
- Large sample size doesn't help with systematic measurement error (Chen et al. 2019)

### Can we measure formant tuning?

*(Not from LPC formant measures alone)*

- 'Formants are easy to measure, resonances not so much' (Whalen et al. 2022)
- **Ground truth test:** Klatt vowels with fixed, known resonances over f0 range
- Resonances based on mean F1/F2 for frequent words by male PoPS singers; re-measured in Praat (5 formants <5 kHz)
- **Result:** LPC estimates are increasingly attracted towards harmonics as f0 rises

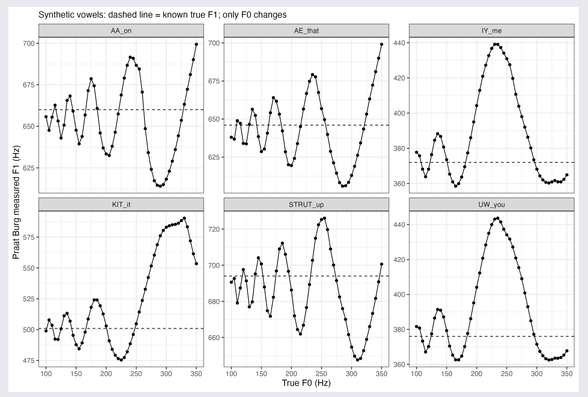



Figure 4. Klatt synthesis provides known underlying resonances: as f0 increases, LPC estimates are attracted towards harmonics.

## Searching for true resonance-harmonic alignment without relying on LPC

Manually inspected a range of high-f0 tokens with multi-note melismas or very prominent harmonics.
Large formant tuning related vowel shifts seem rare, but testing this requires careful manual inspection. In many cases true resonances can not be determined at high f0.

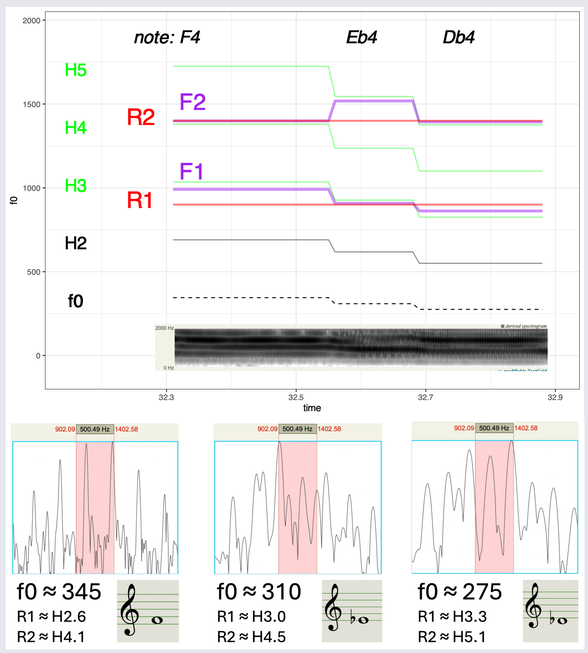
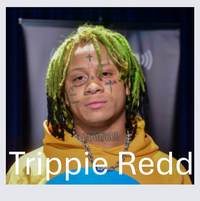



Figure 5. Trippie Redd: 'top' melisma sung across three notes. Spectral evidence consistent with stable vocal-tract resonances (red); LPC estimates (purple) jump towards nearest harmonics (green).

### Shifting f0, stable vowel quality: Harmonic attraction mismeasurement

- Melisma over three notes on 'top'
- Auditory and spectral evidence suggest stable LOT vowel as f0 changes
- As the harmonic grid shifts, different harmonics near the stable resonances
- LPC formant measures jump to nearest harmonic
- Apparent formant-harmonic alignment based on mis-measurement (cf. Fig. 3)
- Spectrogram shows **real accidental alignments**:
  - R1 enhances H3 on the middle note
  - R2 enhances H4/H5 on 1st/3rd notes




### Shifting f0, shifting vowel quality: Possible formant tuning?

- Dylan Storey appears to shift vowel quality to keep a resonance aligned with the same harmonic as f0 moves
- DRESS vowel rhyming scheme: 'question/lemon/feather' (f0=227Hz) 'heavy' (f0=261Hz).
- H8 strongly prominent in all four tokens despite being 270Hz higher.
- 'heavy' raised and fronted to keep F2 close to the new H8? (See Fig. 1)
- Could R2 be stable? *(No, it really shifts!)*
  - Neighbouring H7/H9 amplitudes change in the opposite direction than predicted by a stable R2 with harmonic-attraction account.
- Unresolved: **accidental** alignment vs **intentional** formant tuning

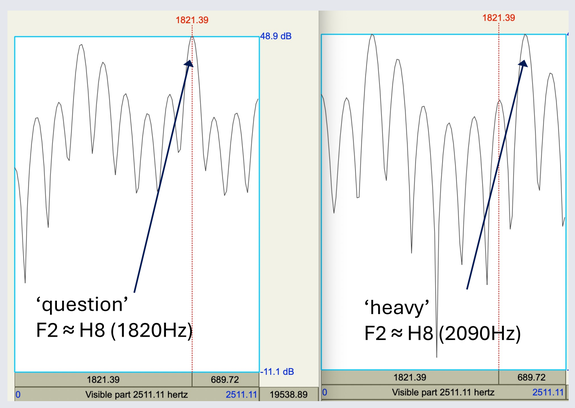



Figure 6. Dylan Storey: spectra of DRESS in 'question' (f0 = 227 Hz, left) and 'heavy' (261 Hz, right). H8 remains prominent as it moves from 1820Hz to 2090Hz. See the corresponding vowel movement in Fig. 1.




## Conclusion

Harmonic attraction makes **formant tuning difficult to measure**
Manual inspection of spectra and auditory analysis needed to distinguish artefacts from genuine alignment
Conspicuous formant tuning appears **rare** in these pop vocals
**Genre** structures singing accents —— Jaw opening on high notes is an important **singing-technique constraint** on vowel quality —— Formant tuning may play a minor role

**References** **Austin (2007).** Jaw Opening in Novice and Experienced Classically Trained Singers. *J. Voice* 21(1) ||| **Barreda (2020).** Fast Track: fast (nearly) automatic formant-tracking using Praat. *Linguistics Vanguard* 7(1). ||| **Chen et al. (2019).** F0-induced formant measurement errors result in biased variabilities. *JASA* 145(5). ||| **Défossez, A. (2021).** Hybrid spectrogram and waveform source separation. ||| **Gibson (2024).** Pop Song English as a supralocal norm. *Lang. in Soc.* 53(3). ||| **Gibson (2026).** Finding Resonance in Speech and Song. *LabPhon20* ||| **Gibson & Bell (2012).** Popular music singing as referee design. Hernández-Campoy & Cutillas-Espinosa (eds.), *Style-Shifting in Public.* ||| **Gibson & Hay (2026).** Perceptual style-shifting across singing and speech: Music activates Pop Song English for NZ listeners. *Lang. & Speech* 69(3). ||| **Herbst et al. (2023).** Acoustical theory of vowel modification strategies in belting. *J. Voice* 39(5). ||| **Joliveau et al. (2004).** Vocal tract resonances in singing: The soprano voice. *JASA* 116(4). ||| **McAuliffe et al. (2026)**. Montreal Forced Aligner and the state of speech-to-text alignment in 2026. ||| **Schutte et al. (2005)**. Resonance strategies revealed in recorded tenor high notes. *Folia Phoniatrica et Logopaedica* 57(5-6) ||| **Story et al. (2025).** Detection of synchronization of the voice source and vocal tract in connected speech. *JASA* 158(3). ||| **Sundberg et al. (2013).** Formant tuning strategies in professional male opera singers. *J. Voice* 27(3). ||| **Trudgill (1983).** *On Dialect: Social and Geographical Perspectives*. ||| **Whalen et al. (2022)**. Formants are easy to measure; resonances, not so much: Lessons from Klatt (1986). *JASA* 152(2).